

Machine Learning

Supervised ML algo. :-

✓✓

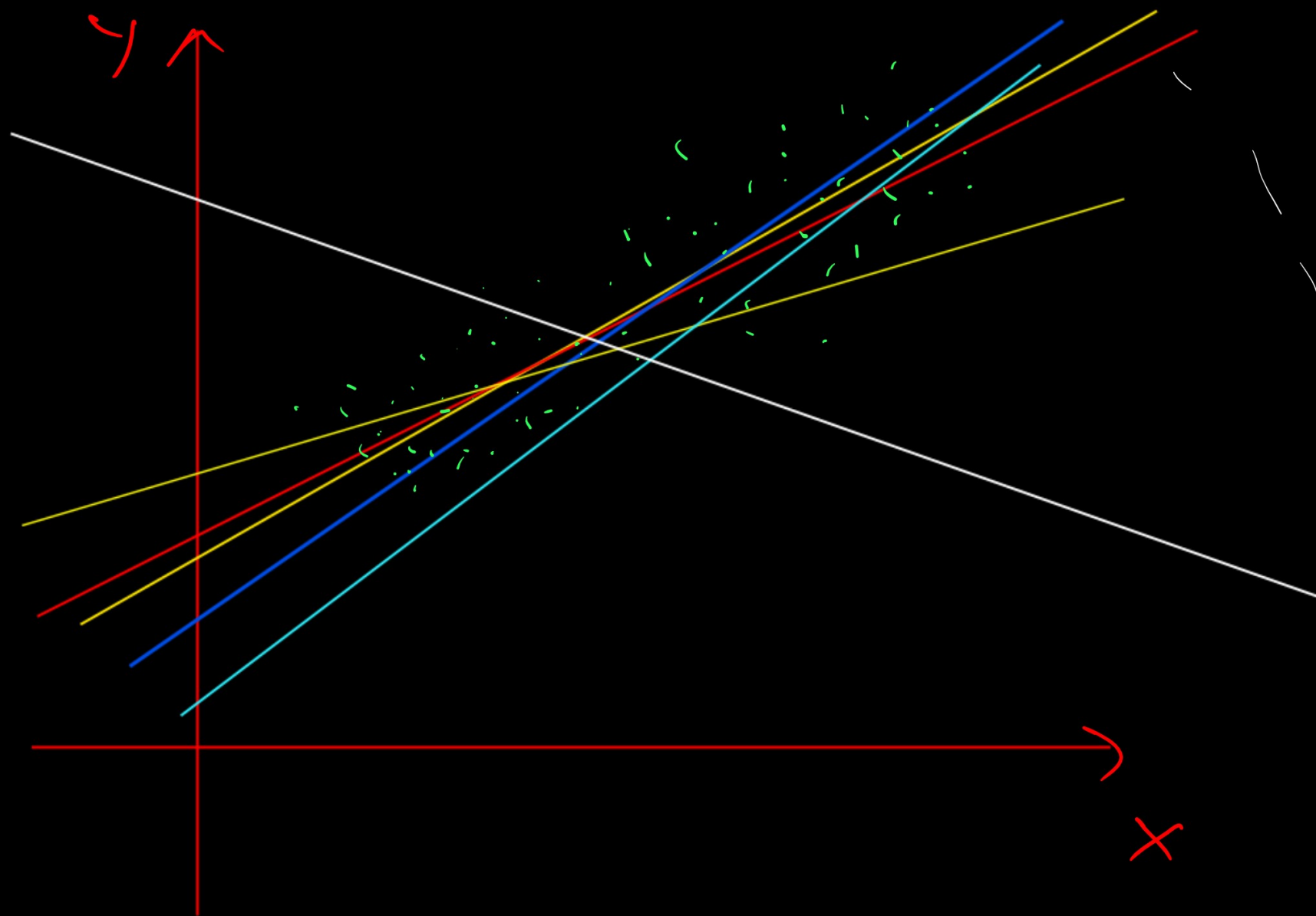
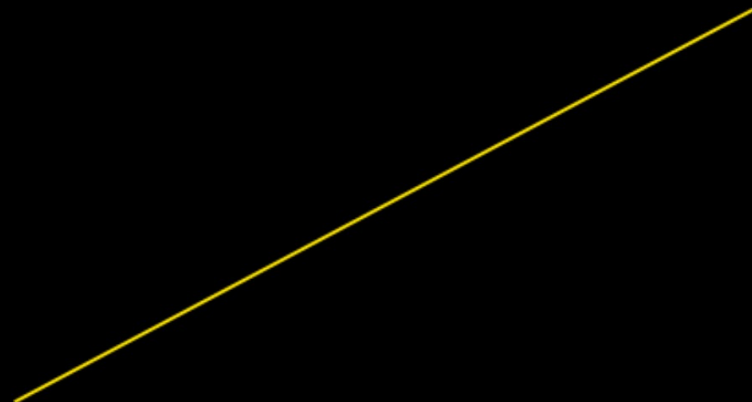
→ regression ✓✓

→ classification ✓✓

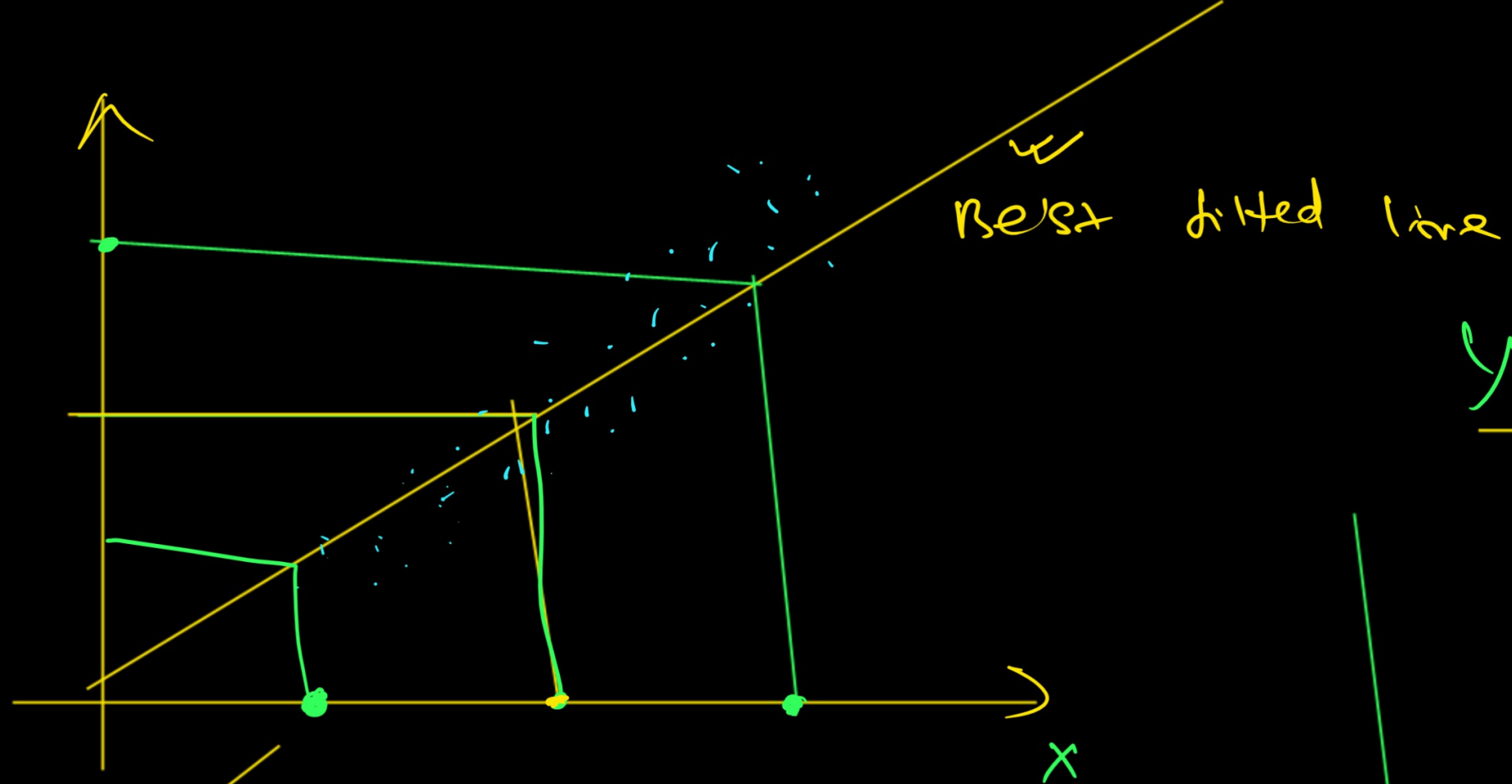
<u>ⓧ</u>	<u>Ⓨ</u>
1	2
2	5
3	7
4	9
<u>5</u>	<u>?</u>

① Linear Regression :-

$$y = mx + c$$



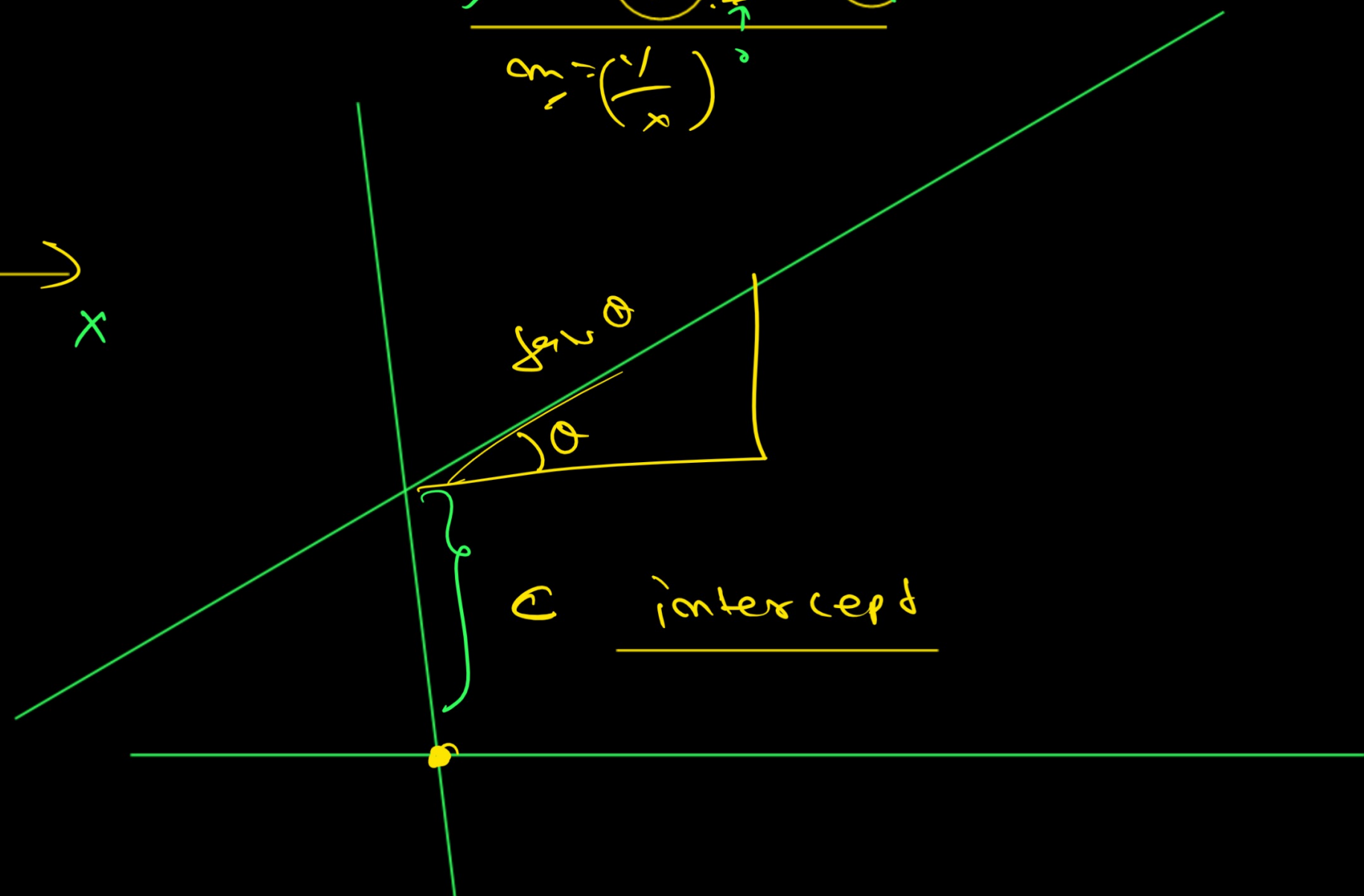
✓



$$y = \textcircled{m}x + \textcircled{c}$$

$m = \left(\frac{1}{x}\right)$

$c = 0$



$$y = 5x + 2$$

$\uparrow$
 m

$\uparrow$
 c

X	Y
→ 1	2
→ 2	3
→ ③	③

$$y = mx + c \quad \leftarrow 0$$

$$2 = m \times 1 + c$$

$$2 = m + c \quad \leftarrow ①$$

$$3 = m \times 2 + c$$

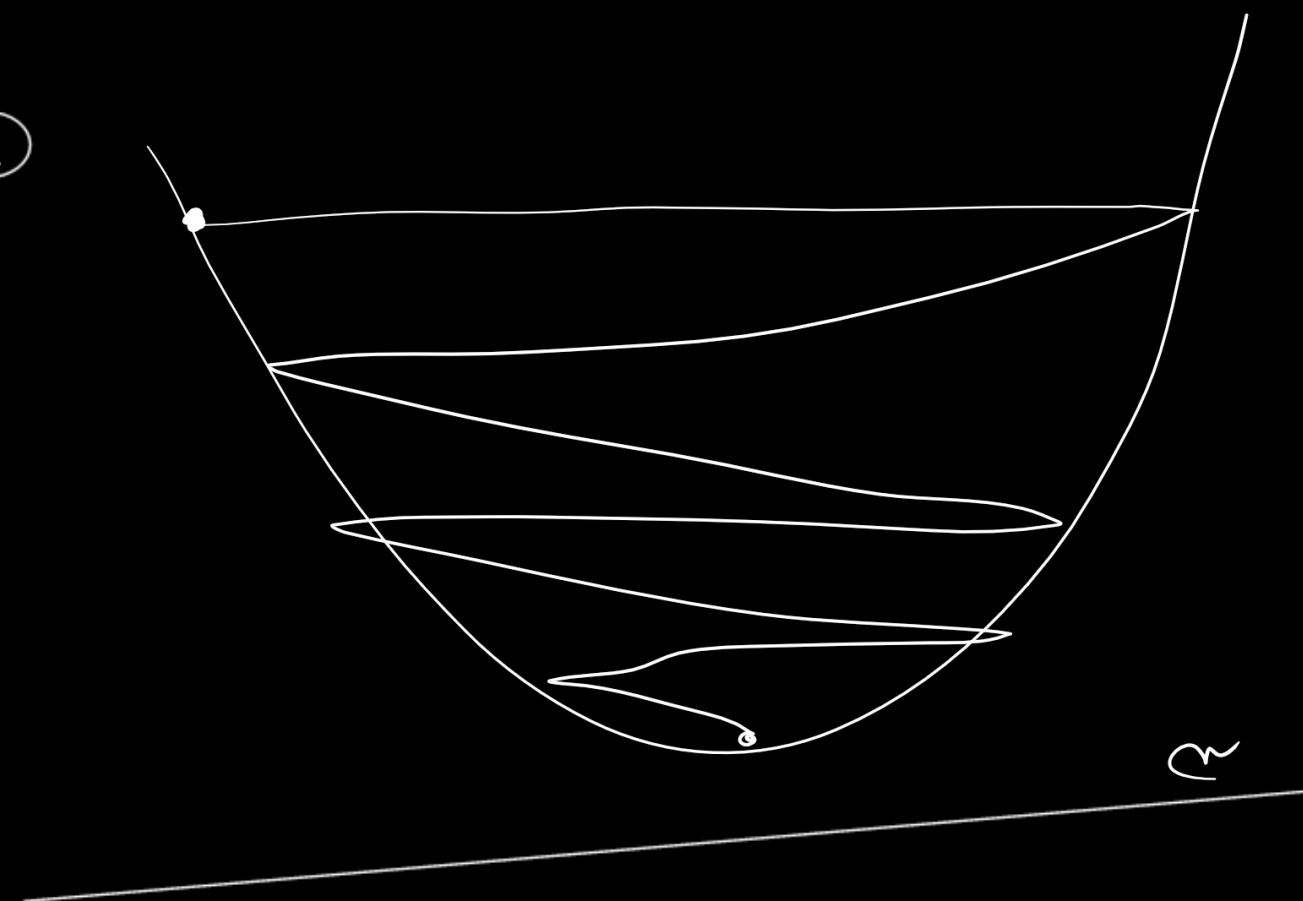
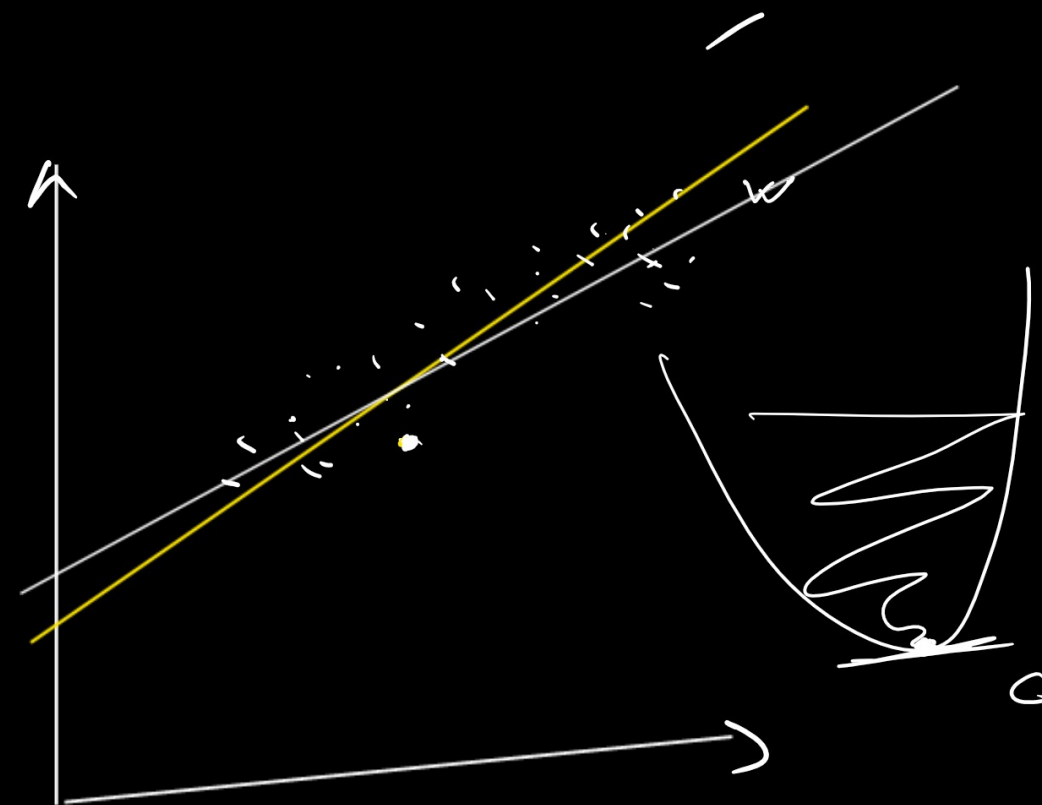
$$3 = 2m + c \quad \leftarrow ②$$

$$y = x + 1$$

$$\hat{y} = 3 + 1 = 4$$

$$\text{Error} = (3 - 4)^2 = (-1)^2 = 1$$

$$m = 1 \quad c = 1$$



$$y = mx + c$$

$$\hat{y} = mx + c$$

x = input

m = slope

$$m = \left(\frac{y}{x} \right)$$

c = intercept

$x=0$

$$y = c$$

y = Expected value

$\hat{y}$ = Predicted value

$$e = (y - \hat{y})$$

Mean squared error :-

$$J(m, c) = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \frac{1}{2n} \sum_{i=1}^n (y - (mx + c))^2$$

$$J = \frac{1}{2n} \sum_{i=1}^n (1 - \sigma_i)^2$$

$$= \frac{1}{2n} \sum_{i=1}^n (1 - (mx + c))^2$$

$$\frac{\partial J}{\partial m} = \frac{\partial}{\partial m} \left(1 - mx - \frac{1}{2} \right) = \frac{1}{2n} \sum_{i=1}^n -2x = -\frac{1}{n} \sum_{i=1}^n x$$

$$\frac{\partial J}{\partial c} = \frac{\partial}{\partial c} (1 - mx - c) = -1$$

$$J = \frac{1}{2n} \int_{-M}^M \left(y - \frac{(mx + c)}{u} \right)^2$$

$$u = (y - mx - c)$$

$$J = \frac{1}{2n} \int_{-M}^M (y)^2$$

$$\frac{\partial J}{\partial m} = \frac{\partial u}{\partial y}$$

$$= \left(\frac{\partial J}{\partial y} \right) \cdot \frac{\partial y}{\partial m}$$

$$= \left(2y \left(\frac{\partial y}{\partial m} \right) \right)$$

$$y = x^2$$

$$\frac{dy}{dx} = (2x)$$

$$\frac{\partial (y - mx - c)}{\partial m} = (-x)$$

$$= 24 \cdot (-x)$$

$$= 2(4 - (mx + c)) \cdot (-x)$$

$$\frac{\partial J}{\partial m}$$

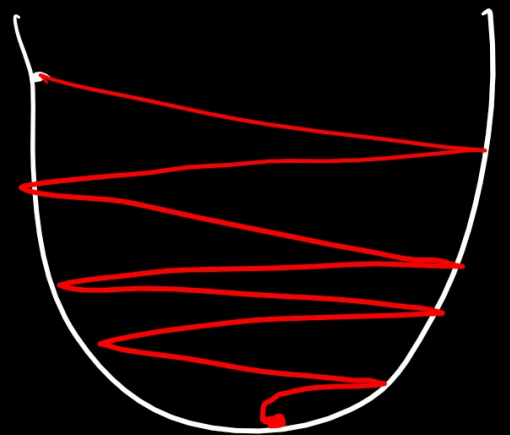
$$\equiv 2(4 - \hat{y}) - (x)$$

$$\left(\frac{\partial J}{\partial m} \right) = \frac{1}{2m} \sum_{i=1}^n 2(4 - \hat{y}_i) (-x_i)$$

$$\equiv -\frac{1}{2m} \sum_{i=1}^n x_i (4 - \hat{y}_i)$$

$$\eta = 0.1, 0.01, 0.001, \dots, 1, 10$$

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$$\cancel{m} = m_0 - \left(\gamma \left(\frac{\cancel{J}}{\cancel{m}} \right) \right)^2 = m_0 - \gamma^2 \left(\frac{1}{2^2} \cancel{M}^2 \right) \cancel{x}^2 (1 - \cancel{v}^2)$$

$$\cancel{c} = c_0 - \gamma \left(\frac{\cancel{J}}{\cancel{c}} \right)$$

$$\frac{\cancel{J}}{\cancel{c}}$$

$$J = \frac{1}{2^2} \cancel{M}^2 (1 - \cancel{v}^2)^2$$

$$J = \frac{1}{2^2} \cancel{M}^2 \left(1 - \frac{\cancel{v}^2 (m + c)}{\gamma} \right)^2$$

$$J = \frac{1}{2n} \sum_{i=1}^n (y_i)^2$$

$$\begin{aligned} \frac{\partial J}{\partial c} &= \frac{\partial J}{\partial y} \cdot \left(\frac{\partial y}{\partial c} \right) = 2y \cdot \frac{\partial y}{\partial c} \\ &= 2y \cdot \frac{\partial (y - \cancel{\text{max}} - c)}{\partial c} \\ &= 2y \cdot -1 \end{aligned}$$

$$= -2y$$

$$= -2(y - (\cancel{\text{max}} - c))$$

$$= -2(c - \cancel{y})$$

$$= \frac{1}{2n} \sum_{i=1}^n -2(c - y_i)$$

$$\frac{\partial J}{\partial c} = \left(-\frac{1}{2} \sum_{i=1}^n (y_i - \hat{y}_i) \right)$$

$$c = c_0 + \eta \frac{\partial J}{\partial c}$$

$$c_n = c_0 - \eta \frac{\partial J}{\partial c}$$

$$= c_0 - \eta \left(-\frac{1}{2} \sum_{i=1}^n (y_i - \hat{y}_i) \right)$$

$$c_n = c_0 + \eta \frac{1}{2} \sum_{i=1}^n (y_i - \hat{y}_i)$$

(a_{nn}, c_n)

x	y
1	5 ✓
2	10 ✓

$$m = 0 \quad c = 0$$

$$\gamma = 0.5$$

$$\hat{\gamma} = 0$$

$$m_n = m_0 - \gamma \frac{1}{n} \left(\sum_{i=1}^n x_i (\hat{\gamma}_i - \gamma) \right)$$

$$\hat{\gamma} = 0 \times + 0 = 0$$

$$c_n = c_0 - \gamma \frac{1}{n} \left(\sum_{i=1}^n \right) (\hat{\gamma}_i - \gamma)$$

$$\gamma =$$

$$m_n =$$

$$c_n =$$

$$\frac{1}{n} \left(\sum_{i=1}^n (\hat{\gamma}_i - \gamma) \right)$$

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$$\underline{m_n} = m_0 - \eta \frac{\delta J}{\delta n}$$

$$\underline{c_n} = c_0 - \eta \frac{\delta J}{\delta c}$$